CBSE 100/100 TOPPER MODEL ANSWER GUIDE

Exam ID: **EXAM-MAT-DAI-20261018-664F52** | Subject: **Mathematics** | Target: **100% Zero-Deduction Step Format**

Q.1. Evaluate using suitable identity: (103)^2 = (100 + 3)^2. [1M]

Topper Solution:
(a) 10,609

Marking Scheme: 1 Mark for identity application

Q.2. The product of monomials (-4p) and (7pq) is: [1M]

Topper Solution:
(b) -28p^2q

Marking Scheme: 1 Mark for correct sign and powers

Q.3. Multiply the binomials: (2x + 5) and (4x - 3). Show all steps. [2M]

Topper Solution:
8x^2 + 14x - 15

Marking Scheme: 1 Mark for distributive law, 1 Mark for combining like terms

Q.4. Solve the linear equation: 5x + 9 = 5 + 3x. [2M]

Topper Solution:
x = -2

Marking Scheme: 1 Mark for transposition (5x - 3x = 5 - 9), 1 Mark for final value

Q.5. Find the area of a trapezium whose parallel sides are 12 cm and 20 cm, and the distance between them is 8 cm. [3M]

Topper Solution:
Area = 1/2 * (12 + 20) * 8 = 128 cm²

Marking Scheme: 1 Mark for formula, 1 Mark for substitution, 1 Mark for answer with cm²

Q.6. Simplify using algebraic identity: (2a - 7)^2. [3M]

Topper Solution:
4a^2 - 28a + 49

Marking Scheme: 1 Mark for a^2, 1 Mark for -2ab, 1 Mark for +b^2

Q.7. A rectangular sheet of paper 44 cm x 20 cm is rolled along its length to form a cylinder. Find the volume of the cylinder. (Take pi = 22/7) [5M]

Topper Solution:
Radius r = 7 cm, Height h = 20 cm. Volume = pi * r^2 * h = 3080 cm³

Marking Scheme: 2 Marks for radius, 2 Marks for volume formula, 1 Mark for unit